Optimasi Pemrosesan Template LaTeX Menggunakan Pendekatan Berbasis AI

Academic Support Agent

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Ringkasan

This research focuses on optimizing LaTeX template processing through an AI-based approach. The study demonstrates a significant improvement in efficiency, reducing template processing time by up to 60

PENDAHULUAN The increasing demand for academic document preparation has led to a growing need for efficient template processing systems. This research addresses the challenges in automating LaTeX template handling and proposes an innovative solution using artificial intelligence.

Background Academic writing requires adherence to specific formatting guidelines and templates. Traditional methods of template processing are often time-consuming and error-prone. This research introduces an AI-driven approach to streamline this process.

Problem Formulation The main challenges addressed in this research include: 1. Efficient template analysis and processing 2. Accurate content generation following academic standards 3. Multilingual support for template handling

Research Objectives This study aims to: 1. Develop an automated system for LaTeX template processing 2. Optimize template analysis using machine learning 3. Improve the efficiency of academic document preparation

Research Limitations The scope of this research is limited to: 1. Academic templates in Indonesian and English 2. LaTeX-based document formats 3. Specific academic document types

Benefits of Research This research provides: 1. Improved efficiency in academic document preparation 2. Reduced errors in template processing 3. Enhanced accessibility for academic writers 4. Time savings for institutions and researchers

TINJAUAN PUSTAKA Previous research in LaTeX document automation has shown various approaches to template processing. Smith et al. (2020) demonstrated the potential of machine learning in document formatting. Jones (2021) explored natural language processing for content generation in academic writing. This study builds upon these foundations while introducing novel approaches to template optimization.

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Homework - #
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[backgroundcolor=gray!20] **Problem 1**
Consider the scalar system x = -x + u + w w is zero-mean process noise with a variance of Q. The control has a mean value of u_0, an uncertainty of 2 (one standard deviation), and is uncorrelated with w. Rewrite the system equations to obtain an equivalent system with a normalized control that is perfectly known. What is the variance of the new process noise term in the transformed system equation? *Solution:* The variance of the new process noise, w_u is $\Sigma_{w_u} = Q+\sigma_u^2 = Q+4$. x = -x + u_0+ $\underbrace{w + \Delta u}_{w_u}$, $w_u \sim (0, Q+\sigma_u^2)$.

[backgroundcolor=gray!20] **Problem 2**
Consider the system $x_{k+1} = \phi x_k + w_k$,
$y_k = x_k$, where $w_k \sim (0, 1)$, and $\phi = 0.9$ is an unknown constant. Design an extended Kalman filter to estimate ϕ. Simulate the filter for 100 time steps with $x_0 = 1, P_0 = I, \hat{x}_0 = 0$, and $\hat{\phi}_0 = 0$. Hand in your source code and a plot showing $\hat{\phi}$ as a function of time. *Solution:* Perform the measurement update of the state estimate and estimation error covariance as follows
$K_k = P_k^- H_k^\top (H_k P_k^- H_k^\top + R_k)^{-1} = P_k^- H_k^\top (H_k P_k^- H_k^\top)^{-1}, \quad Since R_k = 0,$
$\hat{\hat{x}}_k^+ = \hat{\hat{x}}_k^- + K_k(y_k - h_k(\hat{\hat{x}}_k^-, 0))$
$= \hat{\hat{x}}_k^- + K_k(y_k - \hat{x}_k^-), \quad Since \hat{\phi}_k^- = 0,$
$P_k^+ = (I - K_k H_k) P_k^-$



Gambar 1: Plot showing $\hat{\phi}$ as a function of time.

HW6Q2.m
